B.Sc. (Entire Computer Science)-I, Level - 4.5 UG Certificate Level					
Sem:	II				
Paper Category:	DSC4-2 **(Major) (Group-II)**				
Paper Name:	Discrete Mathematics	**Paper Code :** G06-0204			
Credit:	02	**Theory:**	2 Hrs./Week		
Marks:	**UA:** 30	**CA:** 20	**Total:** 50		

Course Objective:

1. To familiarize the prospective learners with a mathematical structure that is fundamentally discrete.
2. To introduce sets and functions, solving recurrence relations and different counting principles.
3. To study or describe objects or problems in computer algorithms and programming languages.

Course Outcomes: Upon successful completion of this course, students will be able to

1. To understand the notion of mathematical thinking, mathematical proofs, and algorithmic thinking, and be able to apply them in problem-solving.
2. To understand the basics of combinatorics, and be able to apply the methods from these subjects in problem-solving.
3. To use effective algebraic techniques to analyze basic discrete structures and algorithms.
4. To understand asymptotic notation, and its significance, and be able to use it to analyze asymptotic performance for some basic algorithmic examples.
5. To understand some basic properties of graphs and related discrete structures, and be able to relate these to practical examples. Course Objectives:

Unit I: Relations and Functions:	**No. of Lectures:15**	**Weightage: 12-18 Marks**

Set and Function: Set, subset, power set, Cartesian product, relation, types of relation, void, universal, identity, reflexive, symmetric, equivalence, anti-symmetric, partial ordering, asymmetric, inverse relation, Matrix representation and graphical representation of relation, in degree and out degree, closure of relation, transitive closure, Warshall’s algorithm, Function: Definition of function as relation, domain, co-domain and range of function, injective (one-one) function, surjective function (Onto Function), bijective function, composition of function

Unit II: Recurrence Relations and Counting principles:	**No. of Lectures:15**	**Weightage: 12-18 Marks**

Recurrence Relations: Explicit formula, recursive formula, recurrence relation, Homogeneous Recurrence Relation with constant coefficients, homogeneous solution, Linear Recurrence Relation with constant coefficients, particular solution, total solution.
Counting principles: Cardinality of a set, Pigeonhole principle, Addition principle, Multiplication principle, Inclusive-exclusive principles for two sets & three sets, Problems

Reference Books:

1. Combinatorics - V.Krishnamurthy
2. Discrete Mathematical structure for Computer Science, Alan Doerr and K Levassuer
3. Elements of Discrete Mathematics - C.L.Liu
4. Discrete mathematics & its applications- K. Rosen
5. Discrete Mathematics and applications K. H. Rosen,Tata McGraw Hill Publishing Company